Lecture 18 // Oct. 2 // Further applications of NST

Last time: We proved the linear indep. of $S^3_{1/n}(K)$ $\forall n>0$, $\forall k$.

To understand lin indep in $\mathcal{G}^3_{\mathbb{Z}}$, we needed to understand behavior of r_s-inv. under #

- $r_0(Y_1 \# Y_2) \geq \min\{r_0(Y_1), r_0(Y_2)\}$
- $\pi_1 = 1$ neg. def cob. from $S^3_{\frac{1}{n+1}}(K)$ to $S^3_{\frac{1}{n}}(K)$ implies strict inequality of r_s-invts, giving some partial info.
- and $S^3_{\frac{1}{n}}(K) = \partial \begin{cases} +\text{-def } W, & n>0 \quad \Rightarrow r_0(-Y)=\infty \\ B^3, & n=0 \\ -\text{-def } W, & n<0 \quad \Rightarrow r_0(Y)=\infty \end{cases}$

Today, we'll discuss some further applications of the NST paper.

Definite Bounding

We pose the following q^n:

Question: Given a $\mathbb{Z}HS^3$, what kind of 4-mfds W can it bound?

Eg, restrictions on
- homology of W
- intersection form of W
- ...

One natural version of this question is

Question: Is there a $\mathbb{Z}HS^3$ with no definite bounding?

Ie, it bounds neither a +-def nor --def 4-mfd.

We've seen that $[D_2 Y] \neq 0$ proves that Y cannot bound a -def ($H_1=0$) mfd. We've hinted about how to remove this homological restriction, which can be done by using $\mathbb{Q}$ coeffs for IFH. As long as $b_1(W)=0$, this works. If $b_1(W)\neq 0$, surgering

W along the generator(s) of the b_i will kill them, and with the correct defn of $-$-def, this new W' will be $--$-def too, and we can apply the same tools as before.

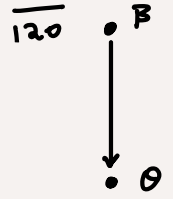
Recall we proved that (even w/o filtration)

$-\Sigma(2,3,5)$ does not bound - def mfd:

Insert example

However, $-\Sigma(2,3,5)$ does bound + def mfd

This is consistant with the fact that if we dualized our cpx, $[D_2]$ would be 0



The point here is that it's not v. easy to use D_2 alone to obstruct a 3-mfd from bounding both types of definite 4-mfds. In fact, it's impossible.

Prop With $\mathbb{Q}$-coeffs, at least one of $[D_2^Y]$ and $[D_2^{-Y}]$ is always 0.

This follows from the relation that $D_2 \circ D_1 \neq 0$.

D_2 class of Y vs D_1 class of $-Y$

We haven't yet said what it is.

Turns out $\exists$ map IFF $\exists$ $V: C_* \to C_{*-4}$ such that $D_2 \circ D_1 = 4(\partial V - V\partial)$

eg, $D_2 \circ D_1$ is nullhomotopic in $\mathbb{Q}$ coeff's, since $\partial V - V\partial$ is nullhx.

This brings us to the following:

Pf (Sketch) If D_1 and D_2 are nontrivial,

eg: • $D_1\alpha \neq 0$ for some cycle α

• $D_2(1)$ is a nonzero hx class

then $(D_2 \circ D_1)(\alpha) = 4(\partial v - v\partial)(\alpha)$

$= 4\partial(v\alpha)$ since α is closed

$\neq 0$

This is why the Q2 is interesting.

Well, what about r_s? Well, it's badly behaved under ori-rev, so there's hope that it can give the answer, which turns out is true.

Thm [NST] $2\Sigma(2,3,5)\#-\Sigma(2,3,11)$ does not bound any def mfd

Recall that $-\Sigma(2,3,6n-1) = S_{1/n}(T_{2,3})$, so

$-\Sigma(2,3,5) = S_{+1}(T_{2,3}) =: Y_1$

$-\Sigma(2,3,11) = S_{1/2}(T_{2,3}) =: Y_2$

Pf First, we show that $-2Y_1 \# Y_2$ does not bound a $-$ def mfd.

For this we use the r_0-invt.

Last time, we constructed a $\pi_1=1$ cob, so $r_0(Y_2) < r_0(Y_1)$.

Rewrite $Y_2 = Y_1 \# Y_1 \# (-2Y_1 \# Y_2)$

By the # formula, $r_0(Y_2) = r_0(Y_1 \# Y_1 \# (-2Y_1 \# Y_2))$

$= \min\{r_0(Y_1), r_0(-2Y_1 \# Y_2)\}$

Since $r_0(Y_2) < r_0(Y_1)$, we must have that

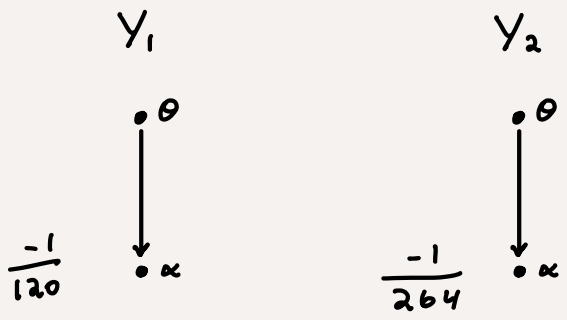
$r_0(-2Y_1 \# Y_2) < r_0(Y_1)$, eg, is finite

Therefore $-2Y_1 \# Y_2$ cannot bound a $-$-def W.

The intuition is that $-$-def picks up minimum.

To show that $-2Y_1 \# Y_2$ does not bound a $+$-def mfd, we can equivalently show that $2Y_1 \# -Y_2$ does not bound a $-$-def mfd.

Consider the unfiltered instanton homology. The essential part of the complex is the flow(s) out of Θ, which for Y_1 and Y_2 as given are:



(the fin family, only diff is that filtr. level of α gets closer to 0

$120 = \gcd?(2\cdot3, 2\cdot5, 3\cdot5)$)

In fact, the complex takes this form for the entire family $-\Sigma(2,3,6n-1)$, the only difference being the filtr. level of α:

$$cs(\alpha) = \frac{-1}{24(6n-1)} \cdot (2_0)$$

Therefore, from the POV of unfiltered instantons, these complexes are the same, so $Y_2 \sim Y$ and $2Y_1 \# -Y_2 \sim 2Y_1 \# -Y_1 \sim Y_1$.
But $[D_2^{Y_1}] \neq 0$, so $[D_2^{2Y_1 \# -Y_2}] \neq 0$. □

Rmk Works like HF d-inv.

Inline: OP = operation
= connect sum

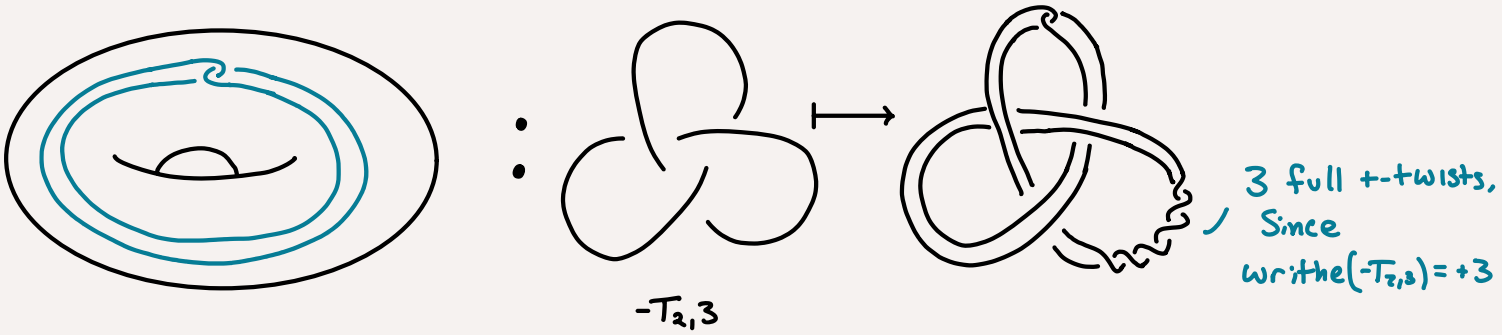
Linear Independence of Whitehead Doubles

Recall that if P is a pattern, the op of satellite with knot K to P is a map from the SM conc. group to itself:
P induces a map $P: \mathcal{C}^{SM} \to \mathcal{C}^{SM}$

What's interesting about this map is that it's not usually a homomorphism. It's an old & deep Q: for diff P's, what is nature of this map?

One particularly interesting P is the Whitehead doubling pattern:

$$Wh: \mathcal{C} \to \mathcal{C}$$



Q Is Wh (weakly) injective?

This Q. is still open and is considered very difficult. To resolve it, one must find some K for which Wh(K) is smoothly slice, but K isn't, or prove no such K exists. For example, it's unknown if $Wh(-T_{2,3})$ is slice (though $-T_{2,3}$ isn't).

Q Is Wh injective?

Since Wh is not a homomorphism, the question of weak injectivity is distinct from that of usual injectivity: does there exist knots K_1, K_2 ST $Wh(K_1)$ is conc. to $Wh(K_2)$ but K_1 is not conc to K_2. This is also still open, as well as the following.

Q Does Wh preserve lin indep?

This was a conjecture of Hedden and Kirk from the 90's, though the question has existed for much longer. Their conjectured answer was yes. At the time it was posed, it was very hard to attack it. In fact, in their paper, they didn't even produce an example of a pair of linearly independent knots whose whitehead doubles are also lin. indep. One surprising fact is that

there are knots K for which $\{Wh(nK)\}_{n=1}^{\infty}$ are lin. indep,

so Wh is really not a hom.
Spreads out the span of what it feeds in

It is still v. hard to attack this problem for two main reasons:

1) Wh(k) is top slice, so no classical alg inv. will work. Thus we need an invariant distinguishing top & sm slices

2) Wh(k) has Seifert genus 1

This means that HFK of Wh(k) is small, eg, it vanishes outside a gr. range which depends on the Seifert genus.

Therefore, obtaining a lin-indep result for an ∞ set is is v difficult, since there's not enough room in HFK to obstruct conc. However, note that in his thesis, Hedden did do it for a finite set for specific k's.

The point here is that its hard to use existing inv's to establish lin indep of an ∞-fam. So what can we do?

Lets translate this q into a into a q about 3-mfds, and ask about hyp cob instead.

Recall the def^n of a double branched cover.

Def A map $f: X^n \to Y^n$ is a branched covering if $\exists\ B^{n-2} \subseteq Y$ st $f: X \setminus f^{-1}(B) \to Y \setminus B$ is a covering map & f is 1:1 on branch locus

B := branch set $f^{-1}(B)$:= branch locus

The double branched cover of a knot $K \subset S^3$ is a a 3-manifold $\Sigma_2(k)$ and covering map: $f: \Sigma_2(k) \to S^3$ such that $f_*: \pi_1($

$$\Sigma_2 : \mathcal{C} \longrightarrow \Theta^3_{\mathbb{Z}/2}$$
$$k \longmapsto \Sigma_2(k)$$

"the" DBC means $\pi_1(Y - B) \to \mathbb{Z}/2$
$\mu_B \longmapsto 1$

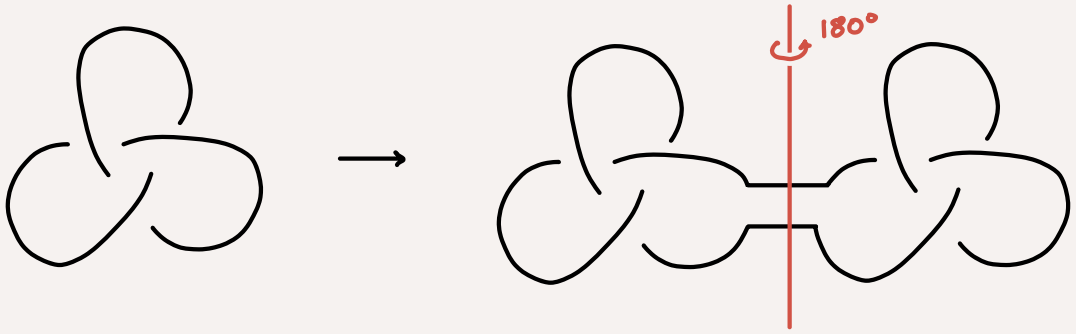
Turns out this is well-defined hom, to $\Theta^3_{\mathbb{Z}/2}$

The point here is that taking the branched cover of a conc gives a cob, so obstructing such a cob obstructs conc. We have methods for this via the r0/rs invt w/ $\mathbb{Z}/2$ coeffs!

Rmk We have (sloppily) kind of been ignoring signs anyways, so using $\mathbb{Z}/2$ instead of $\mathbb{Z}$ shouldn't be too much of a change

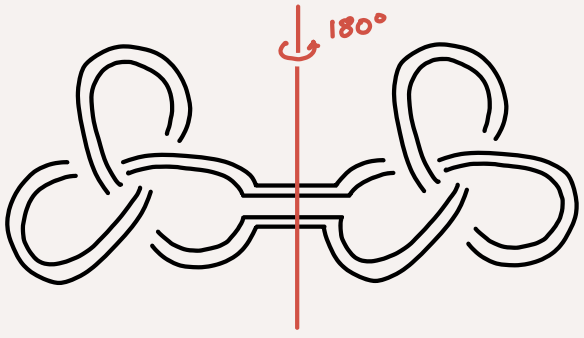
Using Σ_2 is esp. convenient for Whitehead doubles since $\Sigma_2(Wh(k)) = S^3_{\frac{1}{2}}(k \# k)$: Given $K \subset S^3$, first take $K \# K^{(r)}$.

Note that there is a symmetry of S^3 which fixes $K \# K^{(r)}$ setwise given by 180° rotation about the shown axis



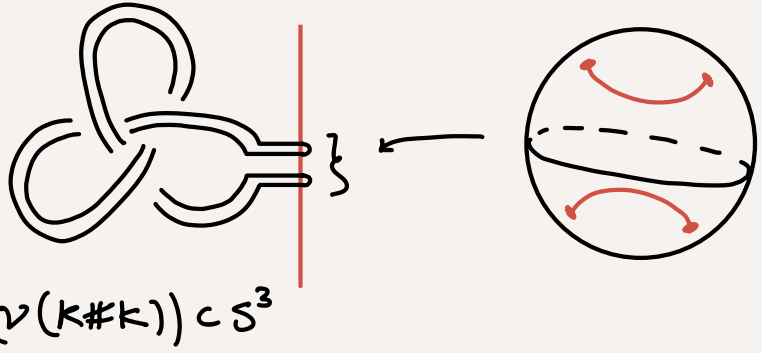
$K \subset S^3$ $\qquad$ $K \# K^{(r)} \subset S^3$

Now, take a tubular nbhd of $K \# K^{(r)}$ and quotient by this rotation.



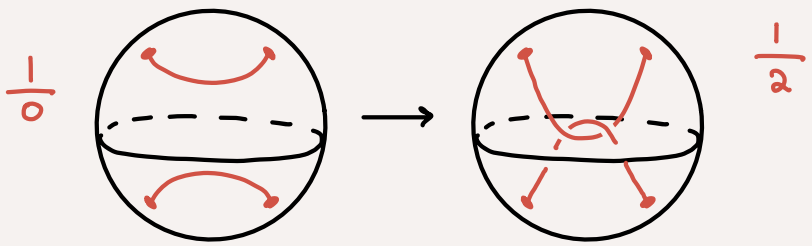
$\nu(K \# K^{(r)}) \subset S^3$

P | 2:1, via 180° rotation across axis

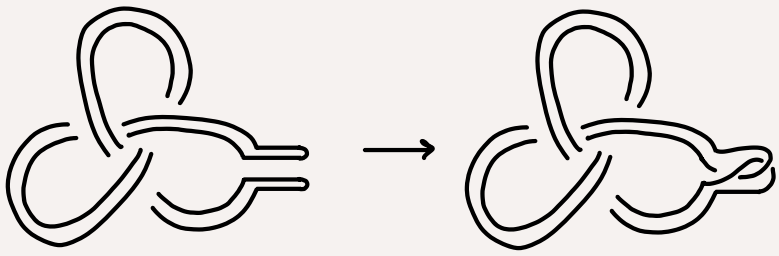


$P(\nu(K\#K))$ is a ball which intersects the branch locus in two boundary-parallel arcs. Before the quotient, surgery on $K\#K$ corresponded to removing $\nu(K\#K)$ and regluing by some map from the bdry torus to itself.

However, in this branch locus, surgery corresponds to ripping out this ball and regluing it by some twist, or, equivalently, a different tangle. Since this ball contains the branch locus, this surgery changes the locus, which changes the branch set upstairs. This is called "rational tangle replacement", where the surgery coeff. p/q determines the tangle, and vice versa. For the Whitehead double, the corr. tangle rep. is



and the branch locus becomes



Note that $S_{1/2}(k\#k)$ is a $\mathbb{Z}/4\mathbb{Z}$-HS^3.

Now we've found a Q about lin indep into a Q about distinguishing $\mathbb{Z}/4\mathbb{Z}$-HS^3's in $\Theta^3_{\mathbb{Z}/2}$, which is the sort of Q we know how to answer. We should be able to calculate some of the IFH invts coming from mfds like $S_{1/2}(k\#k)$.

In light of the full conjecture, this thm is not super impressive—but it is the first major progress.

Thm [NST] $Wh(T_{2,2n+1})$ are lin indep.

There are other families to which this argument can be applied, but we'll outline the proof for the $T_{2,2n+1}$ torus knots for simplicity.

Pf Define $Y_n := S_{1/2}(T_{2,2n+1} \# T_{2,2n+1})$

Then $Y_n = \Sigma_2(Wh(T_{2,2n+1}))$

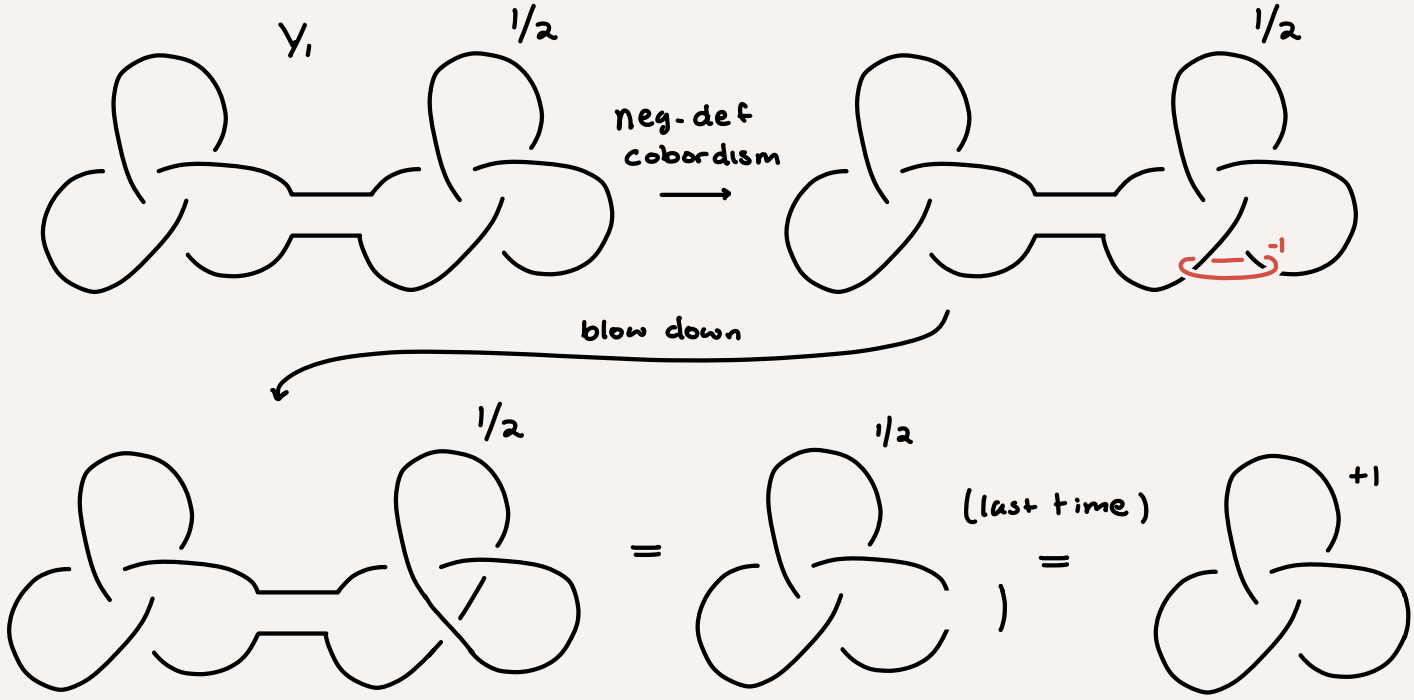
It suffices to show that the $\{Y_n\}_{n=1}^{\infty}$ are lin. indep in $\Theta^3_{\mathbb{Z}/2}$. We'll use the general alg. thm from last time, so we need to know:

2) $\infty = r_0(-Y_1) = r_0(-Y_2) = \cdots$

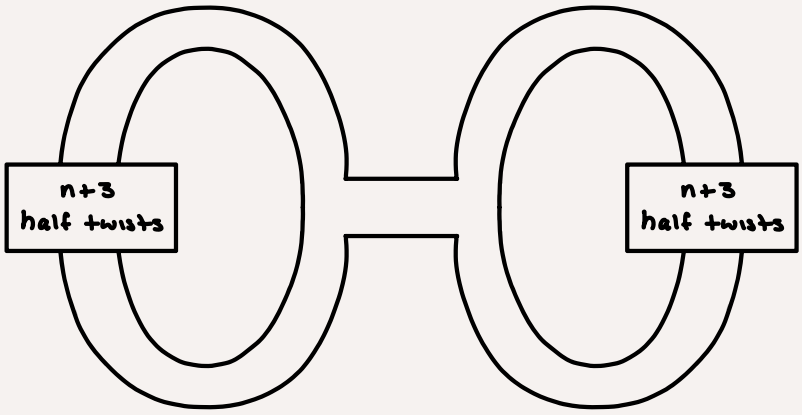
This is true since Y_n is $-\frac{1}{2}$ surgery on k.

1) $\infty > r_0(Y_1) > r_0(Y_2) > \cdots$

We'll similarly construct a neg-def cob from Y_n to Y_{n+1}

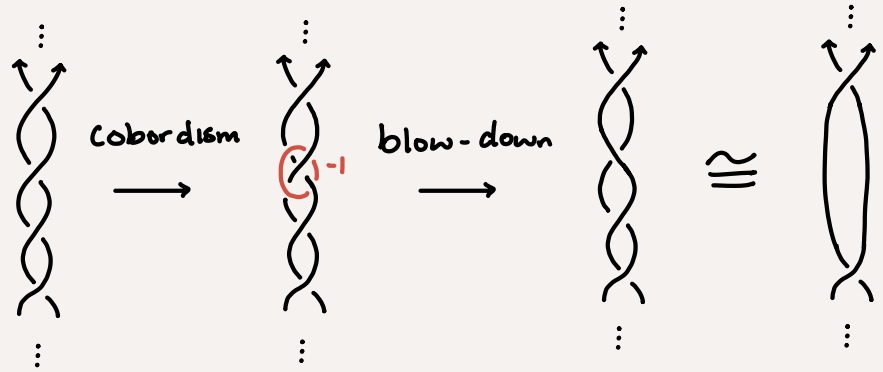


More generally, the picture is



$T_{2n+3} \# T_{2n+3}$

where the cobordism handle attachment and subsequent blowdown decrease the number of half twists on each side by 2:



This gives a neg-def cob from $T_{2n+3} \# T_{2n+3}$ to $T_{2n+1} \# T_{2n+1}$.

To finish our discussion of the NST paper, we'll address some open Q's they mention.

Rationality of CS

Q Are the critical values of the CS functional always rational?

You may have noticed that, at least in our examples, $CS(cpl) = \frac{1}{\#}$. The issue is that the only things we've computed CS for are Brieskorn Spheres and Graph Mfds, all of which have rational CS values. However, the answer to this Q is prob. no.

In the NST paper they propose a counterexample — there's actually there are many diff counterexamples, some of which we expect aren't rational. The primary two are $S^3_{1/2}(-5_2)$ and $S^3_{1/2}(4_1)$.

The CS $\int$'s can be numerically approximated, and they don't look rational, but this isn't an obstructive p.f.

For -5_2, r_0 is believed to be non-rational, and they present consequences of this, if true.

Spse you have a class of 3mds all of which have rational CS values. In the tensor prod cpx for $\#_i$ these must have rational CS, so the rs invt is rational, so there cannot be any 3-mfld in this span with irrational CS value.

A few months ago, N showed that $-S^2$ is not in the span of graph manifolds, which negatively answered the open Q of whether the span of all graph mfds is $\Theta^3_{\mathbb{Z}}$.